

26) Let A, B and C be sets. Show that $(A-B)-C = (A-C)-(B-C)$

Solution:

$$\begin{aligned}
 (A-C) &= (B-C) = (A \cap \bar{C}) - (B \cap \bar{C}) \quad (\text{Prob 21(a)}) \\
 &= (A \cap \bar{C}) \cap \overline{(B \cap \bar{C})} \quad (\text{Prob 21(a)}) \\
 &= (A \cap \bar{C}) \cap (\bar{B} \cup \bar{\bar{C}}) \quad (\text{1st De Morgan's Law}) \\
 &= (A \cap \bar{C}) \cap (\bar{B} \cup C) \quad (\text{Complementation Law}) \\
 &= A \cap (\bar{C} \cap (\bar{B} \cup C)) \quad (\text{2nd Associative Law}) \\
 &= A \cap ((\bar{C} \cap \bar{B}) \cup (\bar{C} \cap C)) \quad (\text{2nd Distributive Law}) \\
 &= A \cap ((\bar{C} \cap \bar{B}) \cup (\bar{C} \cap \bar{\bar{C}})) \quad (\text{2nd Commutative Law}) \\
 &= A \cap ((\bar{C} \cap \bar{B}) \cup \emptyset) \quad (\text{2nd Complement Law}) \\
 &= A \cap (\bar{C} \cap \bar{B}) \quad (\text{2nd Identity Law}) \\
 &= A \cap (\bar{B} \cap \bar{C}) \quad (\text{2nd Commutative Law}) \\
 &= (A \cap \bar{B}) \cap \bar{C} \quad (\text{2nd Associative Law}) \\
 &= (A-B) \cap \bar{C} \quad (\text{Prob 21(a)}) \\
 &= (A-B)-C \quad (\text{Prob 21(a)}) \quad \square
 \end{aligned}$$

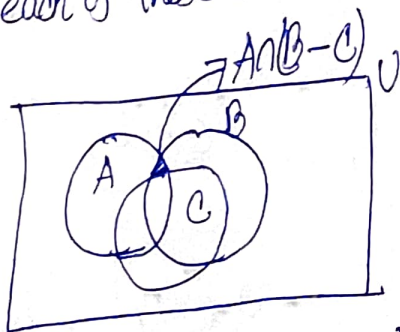
27) Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$.

Find

a) $A \cap B \cap C = \{4, 6\}$ b) $A \cup B \cup C = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 c) $(A \cup B) \cap C = \{4, 5, 6, 8, 10\}$
 d) $(A \cap B) \cup C = \{0, 2, 4, 5, 6, 7, 8, 9, 10\}$

29) Draw the Venn diagrams for each of these combinations of the sets A, B and C.

a) $A \cap (B-C)$



b) $(A \cap B) \cup (A \cap C)$

